

Navigating transient content: PFC caching approach for NDN-based IoT networks

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ABSTRACT

The emergence of Internet-of-Things (IoT) has revolutionized communication among devices. IoT devices autonomously collect and disseminate contents to end-users via network routers. There is growing interest in integrating IoT communications with Named Data Networking (NDN) architecture to retrieve and distribute content efficiently. The content caching characteristics of NDN are pivotal in improving Quality-of-Service (QoS) for IoT. However, unlike multimedia content traffic, which tends to remain static, IoT-generated content is inherently transient in nature, and each content has a finite lifespan. As a result, without efficient caching solutions for IoT contents, the network efficiency and user experience would be degraded. Existing caching approaches often overlook the importance of IoT content freshness, its access pattern and the position of routers during content placement decisions in the IoT networks. In this paper, a novel Popularity and Freshness-based Caching (PFC) scheme has been proposed that aims to strategically cache popular and fresh IoT contents on routers located close to the end-user devices. In the proposed solution, the popularity of content is determined using the request history queue deployed on all network routers. For efficient caching decisions, the hop count metric favors routers in close proximity to end-users. Rigorous simulations with realistic network parameters are performed on the realistic IoT network topologies. The simulation results demonstrate that the PFC approach outperforms existing state-of-the-art caching approaches (LCE, LCC, Consumer-Driven, Consumer-Cache, etc.) on several performance parameters: cache hit ratio, network delay, hop count, network traffic, and energy consumption. This makes the PFC caching approach well-suited for NDN-based IoT networks by enabling efficient content caching decisions.

1. Introduction

Internet of Things (IoT) has emerged as a network of “Things” that integrate sensors, software, actuators and other technologies. These devices are aimed to connect and exchange content with other devices over the Internet. Presently, there is rapid growth in the realization of IoT applications such as smart cities, healthcare, transportation, smart home appliances, hazardous places, etc. [1]. These applications form a connected system that enables automation, data sharing, and intelligent features. IoT devices are generally resource-constrained and have limited computational and memory resources. Additionally, due to the sharp increase in the number of IoT devices, the IoT has begun to experience following issues: (1) Current Internet Protocol (IP)-based Internet architecture cannot seamlessly serve the needs of IoT due to its host-centric nature [2], (2) The load from IoT devices need to be reduced

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to improve their lifetime, and (3) IoT applications generate transient contents that need to be managed efficiently for improved network performance [3,4]. The load from IoT devices refers to the computational, communication, and energy requirements for data sensing, processing, storage, and transmission. Excessive load can lead to faster depletion of battery life that would reduce their operational lifespan.

Many IoT applications, such as weather forecasting, water quality monitoring, remote patient health observation, and industrial automation and control, generate transient content [5]. The transient contents are time-sensitive and become no longer useful (valid) once their lifetime expires. The freshness of the transient content exchanged between devices expresses how recently the content has been created and remains valid for use in the network. In IP-based Internet architecture, the content is accessed from a specific node based on the IP address. However, in IoT applications, users prefer to access the required content from any node having a copy of the requested content instead of searching for the specific node by IP address. Therefore, IP-based architecture suffers from increased network traffic and excessive usage of network resources because the content requests are still forwarded to the specific node/IoT device [6,7]. This is due to the host-centric characteristic of the IP-based Internet architecture, which is not inherently designed for efficient content distribution and caching operations. In addition, the traditional IP-based Internet is not implicitly designed for transient contents and presents scalability challenges, specifically in large-scale IoT deployments. As a result, the conventional IP-based Internet architecture is less suitable for the IoT environment [8].

Towards this, Information-Centric Networking (ICN) offers a convincing alternative to conventional IP-based architectures for IoT applications [9,10]. ICN decouples content from the location and provides secure data objects. Named Data Networking (NDN) and Content-Centric Networking (CCN) are two prominent implementations of ICN. NDN extends the concepts of CCN and focuses more on deployment and integration with existing infrastructure. Due to this, NDN architecture is widely deployed in various IoT applications, video streaming, and distributed content delivery. The ICN framework facilitates the Internet infrastructure transition from the host-centric to content-centric paradigm. This approach simplifies data management in IoT environments by associating content with unique hierarchical names and ensures seamless accessibility of IoT content. ICN offers the in-network content caching characteristic that is considered to be the primary solution for managing the substantial volume of contents produced by IoT devices [11]. Caching is an inherent feature of ICN that reduces communication delay and network traffic while improving resource accessibility [9]. This feature is essential for resource-constrained IoT devices because the content produced by an IoT device can be cached and fetched from the in-network routers by deploying caching decision policies. This approach increases content availability even when an IoT device is not connected to the Internet and enhances battery lifetime for IoT devices. Hence, it is important to ensure that the implemented in-network content caching approach is efficient and has the ability to intelligently manage the content distribution.

The ICN caching strategy performs content placement and replacement decisions. The content placement policy identifies the suitable network router and the content for caching, while the replacement policy specifies which content should be evicted from the cache when it reaches maximum capacity. As a result, a comprehensive analysis of these two policies is crucial to ensure that they meet IoT network requirements. Many previous research works have partially addressed the issue of content freshness during caching, and the handling of transient IoT contents has not been considered [12]. Furthermore, it is worth noting that users do not request all contents uniformly. Generally, a few contents are requested more frequently than others from the content catalogue [13]. Hence, there is a need for an efficient caching solution that considers the freshness, popularity, and attributes of intermediary routers while making content caching selections in the IoT context.

The proposed work aims to optimize the utilization of the available in-network cache capacity of NDN-based IoT networks. A novel content Popularity and Freshness-based Caching (PFC) Scheme has been proposed to maximize the network hit ratio and decrease average traffic. When making caching decisions, the proposed PFC scheme considers several key factors, such as content popularity, proximity to the IoT device, and remaining lifespan of transient content. Implementing strategic caching decisions increases the probability of accessing the contents from nearby routers, resulting in a higher cache hit rate, lower network traffic and load on IoT devices. The PFC scheme calculates a threshold value by considering the content freshness and distance of the router from the IoT device. The threshold value is compared with the content popularity parameter for the caching decision. Using the proposed heuristics, the PFC scheme places the popular and fresh content in close proximity to end-users. The contributions of this paper are summarized as follows:

1. This paper thoroughly analyzes different caching strategies developed for transient contents in ICN/NDN-based IoT networks. It highlights the limitations of existing research efforts and identifies potential for enhancements.
2. A novel content Popularity and Freshness-based Caching (PFC) Scheme has been introduced for NDN-based IoT networks. The approach makes informed caching decisions by considering several key factors such as content popularity, distribution of Interest message generation, content freshness, and distance parameters.
3. Extensive simulations are conducted on realistic IoT-based network topologies to demonstrate the efficiency of PFC scheme using ndnSIM simulator. Simulation results demonstrate that the proposed caching strategy has significantly enhanced network performance and outperformed existing peer approaches in terms of hit ratio, delay, hop count, network traffic, and energy consumption.

The remaining paper is structured as follows. The next section discusses the literature review on IoT, ICN, and content caching approaches. Section 3 explores the system architecture and fusion of IoT with ICN. The proposed PFC scheme is elaborated in Section 4, and the performance evaluation is conducted in Section 5. The findings of the study are concluded in Section 6.

2. Literature review

The Internet of Things (IoT) aims to gather data autonomously without human intervention. The concept of IoT has advanced beyond mere data collection to offer services for information sharing, analysis, and communication [14]. In the beginning, the IoT has been envisioned with a cloud-centric perspective for global deployment [15]. In this approach, the end-users access the content from IoT devices, which is facilitated by middleware (cloud servers) that provides storage and computation tools [16]. However, frequent content access from numerous IoT devices for every end-user request resulted in higher content access delays and increased network congestion. The IoT devices are constrained by finite computational capabilities and energy reserves, which restrict their ability to perform extensive computations and operate for extended working durations. Furthermore, many IoT applications generate transient contents that need to be delivered to end-users promptly to maintain freshness. Due to this, the content freshness heavily relies on the computation capabilities of the cloud servers and their proximity to end-users [17]. This scenario poses significant performance challenges for real-time IoT applications due to the large number of IoT devices in the network and the growing volume of transient contents [18].

As suggested in various research works, the concept of Information-centric architecture presents a promising improvement in addressing the limitations of cloud-centric architectures [19,20]. Existing research works demonstrate the potential for ICN/NDN to scale and handle the evolving requirements of the IoT, providing real-time services to the current IoT applications. In ICN, in-network routers play a crucial role in retrieving requested content from IoT devices and serving end-users. The routers offer caching capabilities to reduce delay and rapid geographical dispersion of information [21]. With the fast increase in the number of IoT devices and their applications, it is evident from prior research that there is a necessity for moderate to high cache space in IoT networks. The cache space needs to be used efficiently to alleviate the load from IoT devices [22]. Hence, the requirement of a competent cache management mechanism has been observed to utilize the available cache resources efficiently and to adhere to the rigorous Quality-of-Service (QoS) needs of IoT applications [23].

The Leave-Copy-Everywhere (LCE) scheme [24] caches IoT content on every intermediary router between the end-user and the content provider. While this method improves content accessibility across the network, it eventually encounters issues such as a high content replacement rate and poor caching performance. This issue arises because the scheme fails to distinguish between popular and infrequently accessed content, causing popular content to be evicted from the cache by rarely used content. Although various caching strategies are presented for ICN-based IoT networks, many researchers have not considered the transient nature of content during caching decisions in the IoT context [25–27]. For instance, a collaborative caching technique for ICN-based Wireless Sensor Networks (WSN) has been suggested in [28]. This strategy makes caching decisions based on the betweenness centrality of in-network routers and the popularity of contents. The method implements different caching capacities on the routers depending on their location. A probabilistic content placement scheme has been proposed by Gundogan et al. [29] for IoT that classifies the contents into *prompt* and *regular* categories. The scheme caches *prompt* contents with higher priority than the *regular* contents. The technique efficiently stores the contents using random probabilities, where different types of contents are assigned different caching probabilities. One of our prior works [30] proposed an average popularity and distance-based caching (APDC) scheme. It improves the caching performance by computing a dynamic caching threshold value based on content popularity and the distance from the content provider. Using these parameters, the APDC strategy caches popular content near the end-users as the caching threshold tends to decrease with an increase in content popularity and distance from the content provider. However, these approaches [28–30] have not considered the transient nature of IoT contents during caching decisions, which makes them unsuitable for the IoT/WSN environment.

Towards addressing the caching of transient IoT content, Vural et al. proposed a caching scheme that focuses on preventing the frequent retrieval of transient contents from IoT devices [31]. The approach presents a trade-off between caching the content close to the requester and maintaining content freshness. The authors illustrate that placing the IoT content closer to requesters substantially reduces network traffic. However, the approach requires substantial processing resources for caching decisions, posing challenges for IoT networking. A probabilistic caching strategy for IoT called *pCasting* has been introduced in [32] to efficiently store the transient content on wireless IoT devices. The strategy offers autonomous caching decisions by considering the freshness of the content, as well as the energy level and storage capacity of the IoT devices. Since the approach caches the contents in IoT devices and not in the network routers, it causes excessive load on resource constraint IoT devices and thus, does not offer an optimal solution. Meddeb et al. [33] developed the *consumer-cache* strategy that caches the content close to the end-users. Specifically, the scheme caches IoT contents on those routers that are directly connected to the end-users. Thus, the approach works similarly to the LCE caching scheme, if each network router is connected to the end-user device(s) in the IoT environment.

The Lifetime-based Cooperative Caching (LCC) scheme [34] performs caching decisions based on the lifetime of IoT devices and the rate at which content is requested. The approach configures the caching threshold value for each intermediate router by considering its location and the content lifetime. A node caches the IoT content if content lifetime is greater than the threshold. The threshold value tends to decrease while moving towards the edge nodes in the network. Therefore, the nodes near the root have highest caching threshold, and the edge nodes set the smallest threshold value. A Consumer-Driven caching approach is described in [35] where end-user specifies the required freshness of IoT content in the request message. If the cached copy of content does not have the mentioned freshness, it results in cache miss. Then the request message is routed to the upstream router or IoT device that produces the content.

An energy harvesting-based IoT caching solution is suggested in [36] that deploys cache space on the network routers (gateways for IoT devices). The approach presents a dynamic caching threshold adaptation method aimed at optimizing the cache hit ratio. The network routers randomly choose their threshold value and arbitrarily explore for a better caching threshold. The computed

value is compared with the freshness of IoT content, and the placement decisions are taken accordingly. The work described an analytical model based on Markov chains to calculate the ideal interval between consecutive cache updates. However, both caching schemes [34,36] suffer from excessive content redundancy in the network due to the independent caching decisions by network routers.

A Multi-Attribute in-network Caching Decision algorithm (MACD) has been suggested in [37]. The scheme computes a MACD score based on the cache size, hop count, content freshness, and energy levels while making caching decisions. It arbitrarily sets up a threshold value and compares it with the MACD score for content caching. However, the strategy lacks consideration of access frequencies, leading to poor cache utilization. Furthermore, the simulation results are obtained on a custom network topology, which limits the reliability of the simulation results. Wang et al. [38] introduced a caching framework for information-centric IoT and aimed to minimize energy consumption and decrease content transmission latency. The technique considers the remaining energy on the sensor nodes for caching decisions. However, both approaches [37,38] have not considered the content popularity while making caching decisions, which results in the inefficient usage of caching resources.

In a recent study, the authors of [39] leverage the in-network caching to improve content delivery in ICN-IoT. The research introduced a Popularity-Topology-Freshness-based caching (PTF) that focus on content popularity, network topology, and content freshness. The in-network routers took caching decisions based on a ranking metric, derived from these parameters. Additionally, the scheme used a gray model prediction technique to compute cache benefit for the incoming content before caching in dynamic IoT environments. Leira et al. [40] presented a Context-based Cache Admission Policy (ctxCAP) to enhance content availability in mobile NDN. The ctxCAP considered multiple contextual metrics such as content popularity, freshness, proximity, source mobility type, and network density for caching decisions and content replication. The scheme increased the size of the NDN request packet by incorporating several additional fields. It provides a selective content caching mechanism by incorporating node coordinates, mobility prediction, request frequency, and hop count. The authors of [41] introduced the PF-EdgeCache mechanism for NDN to enhance IoT performance by dynamically caching popular and fresh contents at the network edge. The scheme eliminated the need for complex clustering and heavy network management. The performance of the scheme has been evaluated on a large-scale transit-stub topology under diverse IoT workloads and demonstrated superior performance as compared to existing caching strategies. However, the scheme [39,41] may increase computational overhead for routers due to continuous updating and ranking content on the routers. Moreover, the gray model for predictions in [39] may lack precision under volatile conditions and could hinder deployment in IoT. Similarly, the inclusion of multiple fields in NDN packets in [40] and node properties may lead to higher processing and storage overhead. These challenges may impact the scalability of [39–41] strategies in large-scale IoT networks having higher content flows.

The in-network routers implement a cache replacement strategy during caching operations when their cache space becomes full [42]. Various traditional caching schemes have been employed in recent research, such as First In First Out (FIFO), Least Recently Used (LRU), and Least Frequently Used (LFU). In the FIFO replacement strategy, the oldest content is evicted from the cache to place incoming content [43]. The LRU policy replaces the least recently used content in the cache with the new content [12]. This strategy is most frequently used in the IoT domain due to its efficiency [32,34]. On the other hand, the LFU algorithm swaps incoming content in the cache with content that has been accessed the fewest number of times. Generally, in real-world applications, a few contents are accessed more frequently than the others. In such networks, the LFU exhibits better network performance than LRU because of its sensitivity to content access frequencies [44].

The existing literature suggests that an efficient content caching scheme is crucial to improve the QoS in the IoT environment. The caching decision encompasses both content placement and replacement decisions. The content placement decision identifies the suitable content and appropriate in-network router for caching. The replacement decision evicts the low-value content from the cache. Most of the existing caching schemes [32,34,37,38] perform caching decisions based on a few parameters such as content freshness or content popularity. The proposed caching scheme aims to optimize these decisions based on the content popularity, freshness, and the distance of the intermediate router to the IoT device.

3. System architecture and assumptions

3.1. System architecture

The IoT architecture consists of three layers as illustrated in Fig. 1: the perception layer, networking layer, and the application layer [45]. The perception layer contains various IoT devices that collect contents from the physical environment, such as weather sensors, fire detection sensors, and video surveillance cameras. The networking layer serves as the backbone of the IoT architecture and is used to further disseminate the contents. This layer facilitates efficient transmission of contents and communication among the application layer devices. The networking layer consists of base stations, routers, and access points that work based on the ICN architecture in which each content has a unique name. For simplicity, the networking layer devices are referred as routers in the remaining paper. This layer has the caching capability for transient IoT contents to optimize content delivery and reduce delay. The application layer represents the interface between end-users and the perception layer. It consists of devices such as smartphones, tablets, and computers interacting with IoT devices through the networking layer. The end-users can access transient contents generated by the IoT devices for various applications such as smart city management, healthcare, environmental monitoring, and home automation, etc.

The content is delivered to end-users from the networking layer if it has been cached on the in-network device. Otherwise, the content is directly accessed from the IoT devices and cached in the networking layer according to the underlying caching strategy. Thus, IoT devices work as a sink for content requests. The caching mechanism retrieves IoT content locally and minimizes the need to access the content from distant IoT devices every time.

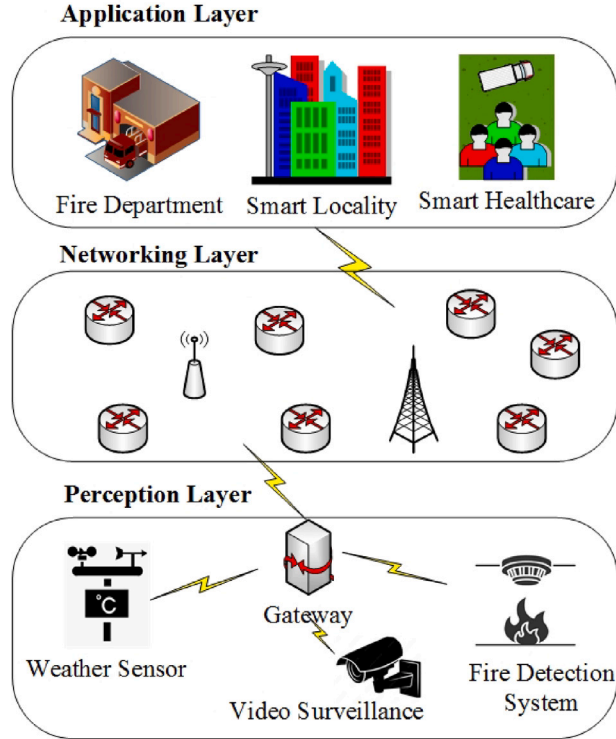


Fig. 1. Illustration of IoT consisting of IoT devices, caching/forwarding devices, and the end-users.

3.2. IoT and ICN fusion: operational landscape

Consider a network topology represented by $G(V, E)$, where V represents the collection of network nodes specified as $V = \{U_1, U_2, \dots, U_{|p|}, R_1, R_2, \dots, R_{|q|}, IoT_1, IoT_2, \dots, IoT_{|r|}\}$. Here, $|p|$, $|q|$, and $|r|$ specify the total number of end-users, in-network routers, and IoT devices in the network, respectively. As shown in Fig. 1, the IoT network architecture consists of three types of network nodes: end-users (U_i), network routers (R_i), and IoT sensor devices. The set E consists of network connections that enable communication between content requesters, routers, and IoT devices. The end-users are connected to the network using nearby routers either through single-hop or multi-hop connections. Therefore, end-users access the contents generated by the IoT devices through network routers. The content payload generated by IoT devices are denoted as $D = \{D_1, D_2, \dots, D_{|D|}\}$. Each content is represented separately as D_j , where $1 \leq j \leq |D|$.

In the framework, Interest messages (I_i) and Content messages (C_i) are the essential elements of the communication paradigm used to retrieve and deliver data. The Interest message (I_i) is a request that is sent from the application layer (end-user) to access content from the IoT device. For instance, an Interest message could request current “temperature data from weather sensor” or “video stream from camera-A” as shown in Fig. 1. A Content message (C_i) is the reply to an Interest message that includes the requested payload (D_i). The content messages (C_i) are forwarded via the networking layer and may traverse several routers (R_x) before reaching the end-user. Traditionally, NDN uses the best-route routing strategy to forward Interest messages in the network [46].

ICN utilizes the Content Stores (CS), Forwarding Information Bases (FIB), and the Pending Interest Tables (PIT) to store and forward Interest and Content messages in the IoT networks [47]. The CS caches transient IoT content. The FIB is used to forward Interest messages towards router’s cache (networking layer) or IoT device (perception layer). The PIT manages pending Interest messages, optimizing content retrieval in IoT scenarios. When an Interest message (I_i) arrives at a router, the router initially searches its CS to determine if it already contains a valid copy of the requested content. If the requested content is present in the CS and is still fresh (i.e., not expired), the router replies with the corresponding Content message (C_i) back to the requester. If the requested content is not found in the CS, then the router searches its PIT. If a matching record is found, then the router aggregates the request with the existing entry and discards the Interest message. Otherwise, it forwards the Interest message to the upstream router/IoT device based on the entry in its FIB. The Content message follows the same path through which the Interest message is received. The router utilizes the entries from PIT to transmit the Content message to the downstream router until it reaches the end-user, who requested the content. Without loss of generality, it is assumed that the size of all transient contents is uniform and the caching capacity of routers $|CS(R_x)|$ is considerably lesser than the size of the content catalogue size $|CS(R_x)| \ll |D|$. Table 1 illustrates the symbols and definitions utilized throughout this paper.

Table 1
Variables definition.

Variable	Definition
R_x	x th- router in the networking layer.
IoT_x	x th- IoT device in the perception layer.
$CS(R_x)$	Content store of R_x .
$ CS(R_x) $	Cache capacity of content store on R_x .
$ D $	Total number of distinct contents generated by IoT devices.
I_i	Interest message.
C_i	Content message.
$N(I_i)$	Name of requested content in Interest message I_i .
$N(C_i)$	Content name in Content message C_i in response to Interest message I_i , i.e. $N(I_i) = N(C_i)$.
$HQ(R_x)$	Request history queue on router R_x .
$HQ^k(R_x)$	k th slot in the request history queue.
$HQ_{C_i}(R_x)$	Occurrence count of $N(C_i)$ in request history queue of R_x .
$Size(HQ(R_x))$	Size of request history queue on R_x .
$MaxSize(HQ(R_x))$	Maximum size of request history queue on R_x .
λ	Interest rate.
$T(R_x)$	Caching threshold on R_x .
α	Popularity skewness parameter in Zipf distribution.
$L(C_i)$	lifetime of C_i .
$Age(C_i)$	Time duration since C_i was created by IoT device.
$F(C_i)$	Freshness of C_i .
$d(I_i)$	Distance traversed by I_i .
$d(C_i)$	Distance traversed by C_i .
λ_{max}	Maximum number of Interest messages generated by an end-user in per unit time.
E_{total}	Total energy required by the network during simulations.

It has been assumed that the Interest message generation pattern follows the Zipf distribution [27,48]. The Zipf distribution offers a mathematical model for measuring the asymmetry of popularity during content request arrival. As illustrated in Eq. (1), a small number of contents are in more demand according to their rank, while the remaining contents are requested less frequently. This distribution is suitable for modeling Interest messages for IoT applications [26]. Eq. (1) defines the popularity of k th- rank content in the content catalogue and α quantifies the skewness of the content popularity. This parameter determines the degree of skewness in the distribution curve, which influences how quickly the request frequency for a content decreases as its rank increases.

$$f(k; \alpha, |D|) = \frac{1/k^\alpha}{\sum_{n=1}^{|D|} (1/n)^\alpha} \quad (1)$$

4. Proposed Popularity and Freshness-based Caching (PFC) scheme

The proposed PFC scheme caches only popular contents with high freshness on the in-network routers. The PFC strategy places these contents near the end-user devices for improved QoS without introducing additional communication overhead. The approach operates autonomously at each router, which ensures no coordination overhead among adjacent routers. Hence, the proposed caching solution mitigates the need for excessive caching operations by implementing a selective caching approach.

The scheme utilizes a request history queue at each network router ($HQ(R_x)$) to determine the popularity of contents. Instead of assigning a fixed size to this queue, its size is adjusted proportionally to the content catalogue size as shown in Eq. (2) to ensure an accurate assessment of the content popularity. If the size of the request history queue is too small, then content popularity estimations may become unreliable due to insufficient entries for analysis. Conversely, an excessively large queue would introduce a high computational delay in identifying the popular content. Therefore, selecting the optimal size of the request history queue is necessary for efficient caching performance.

In Eq. (3), the parameter γ is the regulatory coefficient that correlates between the queue size and content catalogue size. This adaptive sizing strategy ensures the accurate computation of content popularity. When γ is small, fewer Interest messages would be examined to determine content popularity, which may result in inaccurate assessments of content popularity due to high sensitivity. Conversely, a large queue size (corresponding to a high value of γ) can lead to considerable computational delays in identifying popular content. Thus, determining the optimal size for the request history queue is crucial for achieving optimal caching performance.

$$Size(HQ(R_x)) \propto |D| \quad (2)$$

$$Size(HQ(R_x)) = \gamma \times |D| \quad (3)$$

The content freshness parameter also plays an important role in the caching performance because it specifies how recently the content is generated by the IoT device. Caching fresher IoT content leads to improved QoS for end-user devices. Due to this, the proposed caching scheme increases the caching probability with increased content freshness value. This ensures that the content will be valid for a longer duration and serve more content requests than the less fresh IoT content. The freshness of a content is computed using Eq. (4).

$$F(C_i) = \begin{cases} \frac{L(C_i) - \text{Age}(C_i)}{L(C_i)}, & \text{if } \text{Age}(C_i) < L(C_i) \\ 0, & \text{if } \text{Age}(C_i) \geq L(C_i) \end{cases} \quad (4)$$

When $\text{Age}(C_i) \geq L(C_i)$ in Eq. (4), then the freshness of the content C_i would be 0, and the content becomes no longer valid to serve the incoming Interest messages. Thus, the expired content copy is evicted from the CS of the in-network router.

A threshold value ($T(R_x)$) serve to facilitate selective content caching rather than caching every incoming Content message. However, relying on a fixed threshold value may lead to network performance degradation, given the diverse locations of routers within the IoT environment and their observations of different network traffic over time. Hence, it becomes essential to assign threshold values independently to each router based on its geographical location in the IoT-network and the freshness of content. The proposed approach dynamically adjusts the threshold value according to the content freshness value and the normalized distance between the IoT device and the network router. The normalized distance parameter is computed based on the hop count values traversed by the Interest and its corresponding Content message. This comprehensive approach ensures that caching decisions are optimized with respect to content popularity, freshness dynamics, and the geographical distribution of routers within the network. The PFC computes the caching threshold based on Eq. (5) as mentioned below:

$$T(R_x) = \beta \times [(1 - \theta \times F(C_i)) \times (1 - \eta \times \text{Norm}(d(C_i)))] \quad (5)$$

The computation of threshold value involves three coefficients, namely β , θ , and η , which play distinct roles in the caching decision mechanism. The comparison between the value of $T(R_x)$ and the popularity of IoT content determines whether the IoT content will be cached on the intermediate router. Specifically, if the popularity of the IoT content is greater than or equal to $T(R_x)$, indicating a sufficiently high demand or frequency of requests for the content, then the IoT content is cached on the router. Here, β serves as an overall scaling factor, adjusting the magnitude of $T(R_x)$ to align with the recent content popularity determined by the request history queue. θ assigns weight to the freshness of the content and reflects the importance of this parameter in the caching decision process. A higher value of θ indicates a stronger emphasis on prioritizing fresher content for caching. η allocates weight to the normalized distance function $\text{Norm}(d(C_i))$. A higher value of η amplifies the influence of distance parameters on the caching decisions.

The parameter $\text{Norm}(d(C_i))$ is computed using Eq. (6). The value of this parameter tends to increase as the Content message traverses towards the end-user.

$$\text{Norm}(d(C_i)) = \frac{d(C_i)}{d(I_i)} \quad (6)$$

Using Eqs. (4) and (6), the Eq. (7) can be written as follows:

$$T(R_x) = \beta \times \left[\left(1 - \theta \times \frac{L(C_i) - \text{Age}(C_i)}{L(C_i)} \right) \times \left(1 - \eta \times \frac{d(C_i)}{d(I_i)} \right) \right] \quad (7)$$

Determining the optimal values for the coefficients β , θ , and η is crucial for the efficient caching operation in NDN-based IoT networks. However, optimal values for these coefficients are not universally fixed and can vary based on the specific characteristics of the network topology and the nature of the IoT environment. Therefore, empirical evaluation becomes imperative to determine the suitable values for these coefficients in different network scenarios.

4.1. Interest message processing

This section discusses the proposed Interest message processing mechanism for NDN-based IoT networks. On arrival of an Interest message (I_i), each network router (R_x) executes the “Interest message handling procedure” as mentioned below. Here is a detailed outline of the new Interest message processing procedure:

1. Initialization (step 1): When an end-user U_u intends to access a specific IoT content D_i , it generates the corresponding Interest message I_j and transmits it towards the adjacent router in the IoT network. Additionally, the requester initializes the hop count field, $d(I_i)$, in the Interest message with 0.

2. Hop-Count Increment (step 2): On receiving the Interest message I_i , each on-path router R_x increases the distance field $d(I_i)$ by 1 in the Interest message that indicates the number of hops traversed by the message.

3. Request History Queue Processing (steps 3 and 4): When router R_x receives the first Interest message, it initializes the history queue by placing the requested content name ($N(I_i)$) in the first slot. Subsequently, the requested content name is inserted in $HQ(R_x)$ using the First-In-First-Out (FIFO) mechanism. Later, during Content message processing, the router R_x analyzes the ($HQ(R_x)$) to assess the popularity of IoT content.

4. Search CS (step 5): Router R_x searches its CS ($CS(R_x)$) for the matching content D_i . If the content exists, then the “Content message processing” procedure is executed, which is discussed in Section 4.2. Otherwise, traditional Interest message processing is employed by R_x as discussed in Section 3.2.

Interest message processing procedure (U_u, I_i, R_x)

1. U_u prepare I_i , initializes $d(I_i) := 0$ and transmit I_i to R_x .
2. Each on-path router R_x performs: $d(I_i) := d(I_i) + 1$
3. If $HQ^0(R_x) = NULL$,
 - (a) $k := 0$ for R_x .
 - (b) $HQ^k(R_x) := N(I_i)$.
4. Else $k = (k + 1) \bmod \text{MaxSize}(HQ(R_x))$
 - (a) $HQ^k(R_x) := N(I_i)$.
5. Search $CS(R_x)$ for $N(I_i)$. If $N(I_i)$ found then execute “Content message processing” procedure else perform traditional Interest message forwarding based on PIT and FIB.

4.2. Content message processing

In this section, we have discussed the proposed Content message processing mechanism for NDN-based IoT systems. When the content found in the cache of a router (R_x)/IoT device (IoT_r) or the Content message C_i is received by the intermediate router (R_y), then the following procedure is executed.

1. Content Provider Identification (step 1): On arrival of I_i , if the corresponding Content D_i exists in the cache of R_x or I_i reaches the IoT device (IoT_r), then R_x/IOT_r becomes the content provider. The router initiates the “Content message processing” procedure and prepares the Content message C_i with distance parameters as mentioned in step-1 of the Content message processing procedure.

2. Processing on intermediate routers R_y (step 2): When intermediate router R_y receives the incoming Content message C_i , it determines the popularity of the content from its request history queue $HQ_{C_i}(R_y)$ and compute $T(R_y)$ using Eq. (7). If $HQ_{C_i}(R_y) \geq T(R_y)$, then R_y caches the content in its cache ($CS(R_y)$) using LFU-replacement strategy. Finally, router R_y forwards the C_i towards the end-user(s) based on the Pending Interest Table (PIT) entries of router R_y .

Content message processing procedure ($U_u, IoT_r, C_i, R_x, R_y$)

1. R_x/IOT_r prepare C_i with requested payload D_i .
 - (a) Replicate $d(I_i)$ from I_i to C_i .
 - (b) Initializes $d(C_i) := 0$.
2. On intermediate router R_y ,
 - (a) $d(C_i) := d(C_i) + 1$
 - (b) Compute $T(R_y)$ using Eq. (7).
 - (c) If $(HQ_{C_i}(R_y) \geq T(R_y) \text{ or } |CS(R_y)| < \text{MaxSize}(|CS(R_y)|))$, then place C_i in $CS(R_y)$ using LFU replacement.
3. R_y forwards C_i to U_u according to PIT records.

Thus, the proposed PFC caching strategy increases the caching probability for those contents that are popular and has high content freshness. Moreover, the proposed approach cache the identified contents on those network routers that are far from the content provider.

5. Performance evaluation

To assess the performance of the proposed PFC and the peer caching strategies, widespread simulations are conducted using the ndnSIM 1.0 simulator, which is a well-established simulator for NDN research [49]. ndnSIM is developed on the top of the NS-3 network simulator and provides a flexible model to simulate NDN architecture. In this simulator, Zipf distribution-based content requests can be generated to analyze real-world access patterns, and it also provides standard libraries for performance measurement. The performance of the caching schemes has been evaluated on the stub (Abilene-based [50], US26-based [51]), core (SprintLink core network-based topology [52]), and edge-network-based IoT environment [53]. These network topologies consist of five types of nodes: users, routers, servers, gateway, and IoT devices. The simulation topologies are modified versions of the Abilene and US26 network configurations to better reflect the integration of IoT environments. The modifications include the inclusion of IoT devices connected to the server using gateway. The users are distributed across different nodes to simulate diverse geographical locations. Each network topology has 100 IoT devices. The Abilene-based topology comprises 33 network routers, and 133 user devices as shown in Fig. 2. Fig. 3 illustrates the US26-based topology that includes 100 IoT devices, 26 network routers, and 25 users. Similarly, the SprintLink core network and Edge-network topologies have been generated as shown in Figs. 4 and 5. Each network topology has a server and a gateway and the IoT devices are connected to the server using a gateway.

The Interest messages generation follows Poisson distribution, representing random and independent request generation events within the network. This models scenarios where users generate requests at unpredictable intervals. Each end-user generates $[0, \lambda_{\max}]$ Interest messages per unit of time. The Interest messages generated by end-users follow the Zipf distribution between $\alpha = 0.7$ and $\alpha = 0.9$. The Zipf distribution [30] introduces skewness in content popularity, where certain contents are requested more frequently than other contents. The joint consideration of Poisson and Zipf distribution aligns the simulator with real-world user behavior in content consumption.

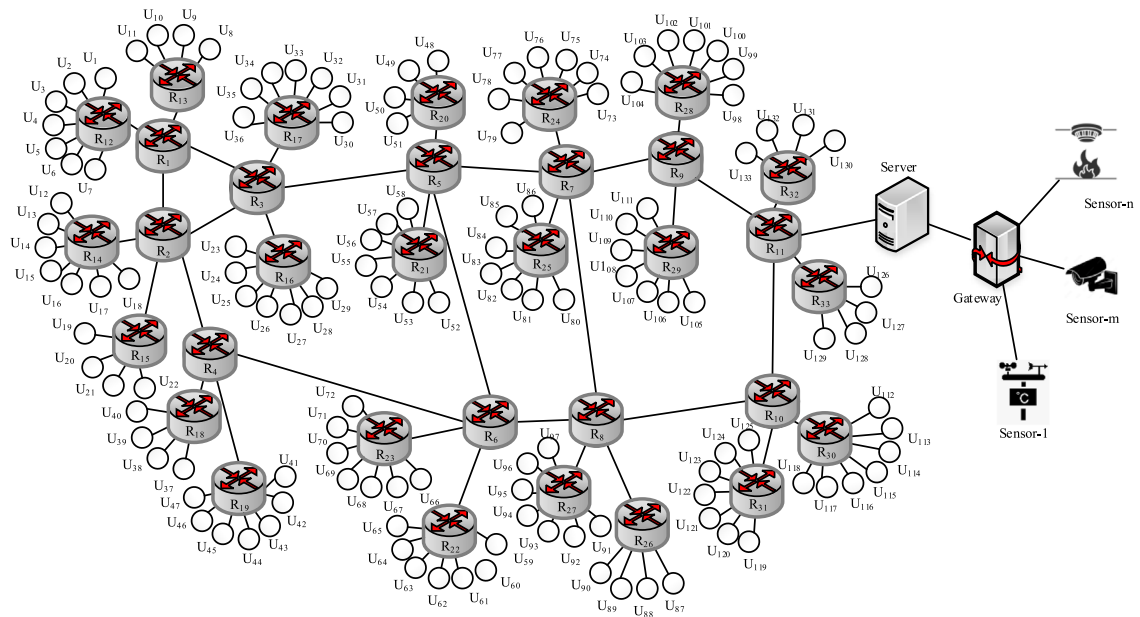


Fig. 2. Abilene-based network topology.

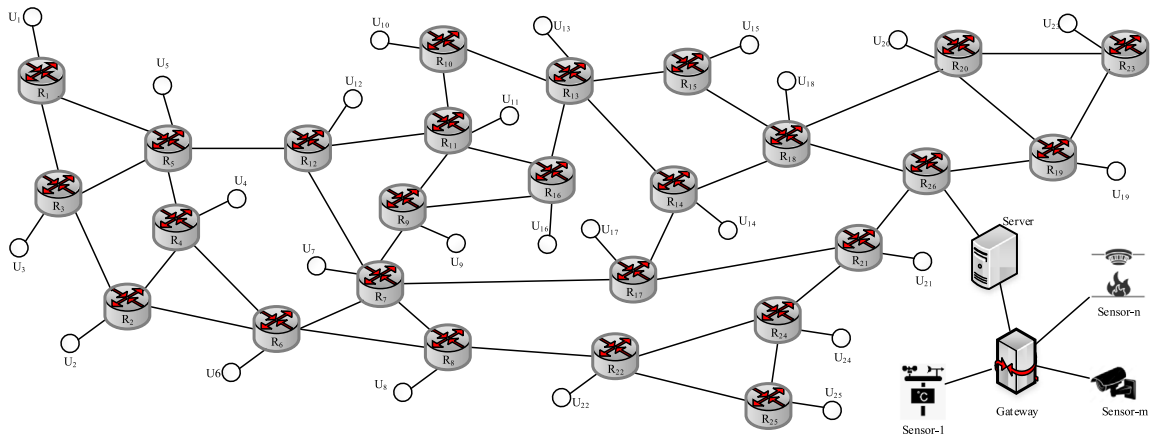


Fig. 3. US26-based network topology.

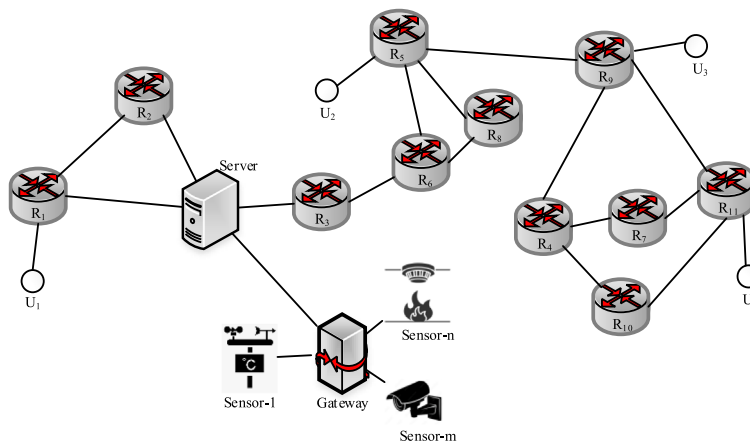


Fig. 4. SprintLink backbone network-based topology.

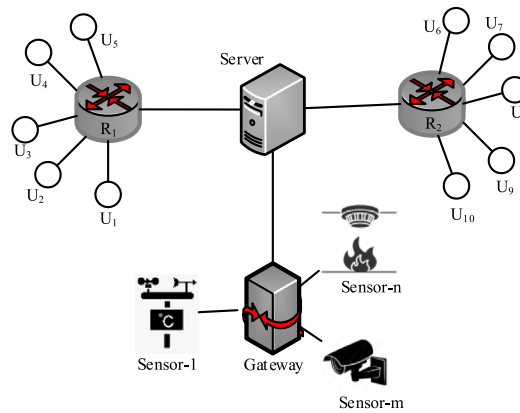


Fig. 5. Edge network-based topology.

5.1. Determination of optimal values of coefficients (β, θ, η) under different characteristics of IoT-networks

We have conducted extensive simulations of the proposed PFC caching scheme on the Abilene-based and US26-based network topologies by systematically varying the values of β , θ , and η with changing the characteristics of IoT networks. The analysis focuses on crucial performance metrics such as cache hit ratio, delay, average hop count, network traffic, and energy consumption. These insights aim to show the influence of β , θ , and η with changing cache size on each router and the content popularity skewness. Tables 2 and 3 summarize the simulation results for different values of caching decision coefficients, cache capacities, and popularity distributions. The caching capacity of network routers is maintained uniformly. To obtain realistic simulation results, the caching capacity of each router has been varied between 20–50, which is 0.4%–1.0% of the content catalogue size (total 5000 transient IoT contents). The IoT devices are deployed within the coverage of a single gateway, and 50 unique contents are produced by each IoT device i.e., a total of 5000 different types of IoT contents. These contents have a lifetime of 4 s, differentiating IoT applications from traditional networking applications. The network links are assigned 10Mbps bandwidth in Abilene-based and US26-based networks. It has been observed that with identical caching capacity, an increase in the value of parameter α leads to improved caching performance due to high skewness in the content popularity distribution. In contrast to $\alpha = 0.7$, when α is set to 0.9, more number of requests are generated for a smaller subset of contents. As the proposed PFC strategy caches popular contents with high probability, it resulted in an improved cache hit ratio.

The simulation results indicate that the cache hit ratio initially increases with θ up to a certain point and after which it decline as θ continues to rise. This trend arises because a lower value of θ promotes the caching of fresher contents in the intermediate routers. However, when $\theta = 0$ and other parameters remain unchanged, the cache hit ratio drops sharply because the threshold value $T(R_x)$ would become high. This prevents relatively fresh contents from being cached in the intermediate routers. Conversely, when θ is excessively high, the value of $T(R_x)$ decreases, which allows less fresh contents (near the end of its lifespan) to be cached. This can lead to the replacement of fresher contents from the cache. Thus, extreme values (low or high) of θ would result in poor network performance. Therefore, an increase in θ is recommended in those networks that have high caching capacity or the administrator is willing to cache those contents also that are near their expiration time.

A similar pattern is observed with coefficient η . An increase in its value reduces $T(R_x)$, enhancing the caching probability at the intermediate routers. However, if η surpasses the optimal value while other parameters remain fixed, the PFC strategy begins content caching operations near the content providers also. Thus, a low value of η is recommended in those networks that have high caching capacities or large path stretches (network diameters) with an aim to cache popular and fresh contents near the edges of the network.

The value of parameter β links the threshold value $T(R_x)$ with the size of the request history queue $MaxSize(HQ(R_x))$. The value of β should scale proportionally with the request history queue size to optimize the caching frequency. If $MaxSize(HQ(R_x))$ increases and the β remains fixed, then more content replacement operations would take place because $HQ_{C_i}(R_x)$ value becomes high in a larger request history queue. Similarly, a low value of β increases the content caching probability and may trigger frequent content replacement operations. In the extreme case where β becomes 0, the PFC scheme behaves similarly to the LCE strategy. Conversely, if β is too high, the resulting high value of $T(R_x)$ prevents caching in the intermediate routers. Thus, β should be increased in those networks that have lesser caching capacity and aim to reduce content replacement operations.

Therefore, the values of β , η , and θ should decrease as the caching capacity increases. This would improve the caching probability in the in-network routers. Table 2 presents performance comparisons across various parameter combinations, showcasing how iterative tuning of these coefficients maximizes caching efficiency. While we have obtained simulation results for different cache sizes and popularity skewness parameters for PFC, the performance of the PFC scheme is compared with existing strategies on simulation parameters outlined in Table 4. The optimal value of the coefficients for Abilene and US26-based topologies under these simulation parameters (mentioned in Table 4) have been highlighted in bold in Tables 2 and 3, respectively. Simulations conducted

Table 2Caching performance for varying coefficient values (β, η, θ) and network characteristics for Abilene-based topology.

Simulation no.	β	η	θ	$ CS(R_x) $	α	Cache hit ratio (%)	Delay (μ s)	Average hop count	Network traffic (KB/s)	Energy consumption (mJ/s)
S-1	2	0	0.1	20	0.7	3.82	115 450	11.71	7737	10 707
S-2	2	0.1	0	20	0.7	3.86	115 267	11.67	7723	10 650
S-3	2	0.5	0.1	20	0.7	3.87	115 192	11.66	7707	10 645
S-4	5	0.1	0.2	20	0.7	8.94	94 984	9.66	6348	8627
S-5	5	0.1	0.5	20	0.7	3.89	115 139	11.65	7701	10 634
S-6	5	0.2	0.2	20	0.7	3.84	115 345	11.68	7727	10 685
S-7	5	0.5	0.2	20	0.7	3.94	114 759	11.63	7672	10 626
S-8	5	0.5	0.5	20	0.7	3.81	115 568	11.72	7744	10 734
S-9	2	0.2	0.1	20	0.9	8.89	95 403	9.71	6404	8703
S-10	5	0.2	0	20	0.9	8.99	94 663	9.61	6346	8625
S-11	5	0.5	0	20	0.9	9.2	93 942	9.56	6294	8569
S-12	5	0.2	0.5	20	0.9	8.88	95 443	9.71	6407	8712
S-13	2	0.5	0.5	30	0.7	4.41	112 991	11.43	7593	10 458
S-14	5	0.2	0.5	30	0.7	4.19	114 086	11.56	7656	10 578
S-15	10	0.2	0.2	30	0.7	2.73	119 323	12.14	7808	10 919
S-16	2	0.1	0.1	30	0.9	9.8	93 061	9.45	6287	8496
S-17	2	0.2	0.5	30	0.9	10.17	92 576	9.38	6234	8421
S-18	5	0.2	0.2	30	0.9	10.22	92 198	9.37	6224	8388
S-19	5	0.5	0.1	30	0.9	9.96	92 942	9.43	6238	8441
S-20	10	0.1	0.5	30	0.9	8.01	98 439	10.03	6459	8778
S-21	5	0.5	0.2	40	0.7	4.5	112 827	11.42	7492	10 353
S-22	10	0.5	0	40	0.7	3.42	116 194	11.8	7754	10 784
S-23	10	0.2	0.1	40	0.7	3.26	117 554	11.95	7804	10 784
S-24	10	0.5	0.1	40	0.7	3.4	116 439	11.82	7756	10 784
S-25	5	0.1	0.1	40	0.9	10.34	92 094	9.36	6212	8383
S-26	5	0.1	0.5	40	0.9	10.48	91 480	9.29	6209	8376
S-27	10	0.1	0.5	40	0.9	8.13	97 263	9.91	6448	8757
S-28	10	0.2	0.1	40	0.9	8.15	96 661	9.85	6443	8731
S-29	10	0.5	0.1	40	0.9	8.67	95 774	9.72	6413	8726
S-30	10	0.5	0.5	40	0.9	8.93	95 238	9.68	6388	8694
S-31	10	0.5	0.2	40	0.9	8.85	95 473	9.72	6408	8725
S-32	10	0.1	0.2	50	0.7	3.71	115 624	11.75	7748	10 780
S-33	10	0.5	0.2	50	0.7	3.82	115 370	11.69	7735	10 685
S-34	10	0	0.5	50	0.9	8.92	95 365	9.68	6389	8697
S-45	10	0.2	0.5	50	0.9	9.17	94 071	9.58	6300	8581
S-36	10	0.5	0.5	50	0.9	9.16	94 481	9.59	6323	8624

on the Abilene, US26-based, SprintLink and Edge network-based topologies revealed slight deviations in optimal coefficient values due to differences in available caching capacity and traffic distribution. The obtained optimal values are summarized in Table 4 for different network topologies. For comparisons, we implemented state-of-the-art LCE [24], Consumer-cache [33], Consumer-Driven [35] caching, LCC [34], and MACD [37] caching schemes. The performance of these existing approaches is obtained and compared with the PFC scheme on various metrics such as hit ratio, network traffic, energy consumption, etc.

5.2. Performance evaluation

This section discusses the simulation results obtained on Abilene-based (Fig. 2), US26-based (Fig. 3), SprintLink (Fig. 4), and Edge network-based topologies (Fig. 5). The performance of the proposed PFC scheme has been compared on several metrics such as cache hit ratio, network delay, hop count, etc.

5.2.1. Performance comparison: average cache hit ratio

When an end-user initiates Interest messages for IoT content, the router examines its cache to determine if the requested content is already stored locally. If a valid copy of the content is found, it is considered a cache hit. Conversely, if the content is not present in the cache, a cache miss occurs, prompting the router to retrieve the IoT content either from the originating IoT device or another router within the network based on the routing strategy. The network cache hit ratio has been computed using Eq. (8).

$$HR = \frac{\sum_{i=1}^{|r|} H(R_i)}{\sum_{i=1}^{|r|} H(R_i) + \sum_{i=1}^{|r|} M(R_i)} \times 100 \quad (8)$$

Here, HR and $|r|$ represent the average network hit ratio obtained in each STU and the total number of routers in the network, respectively. The $\sum_{x=1}^{|X|} H(R_x)$ and $\sum_{x=1}^{|X|} M(R_x)$ represents the sum of cache hit and cache miss count occurred on router $R_i = \{1, 2, \dots, |r|\}$.

Fig. 6 shows the cache hit-ratio results for IoT contents across both Abilene and US26-based network topologies. As depicted in Fig. 6(a), both the Consumer-Driven and LCE schemes demonstrate the lowest cache hit ratios, rendering them unsuitable for deployment in Abilene-based or similar network topologies. The difference highlights the inefficiency of indiscriminately caching content at every router (LCE), leading to cache pollution. Surprisingly, the MACD caching scheme, which incorporates content

Table 3Caching performance for varying coefficient values (β, η, θ) and network characteristics for US26-based topology.

Simulation no.	β	η	θ	$ CS(R_x) $	α	Cache hit ratio (%)	Delay (μ s)	Average hop count	Network traffic (KB/s)	Energy consumption (mJ/s)
S-1	2	0	0.1	20	0.7	5.87	94 104	8.89	1113	1658
S-2	2	0.1	0	20	0.7	7.63	88 934	8.27	1060	1586
S-3	2	0.5	0.1	20	0.7	5.97	93 693	8.85	1111	1651
S-4	5	0.1	0.2	20	0.7	5.5	94 649	9.02	1134	1691
S-5	5	0.1	0.5	20	0.7	5.46	94 704	9.04	1139	1696
S-6	5	0.2	0.2	20	0.7	5.45	94 740	9.18	1142	1698
S-7	5	0.5	0.2	20	0.7	5.56	94 601	8.98	1128	1684
S-8	5	0.5	0.5	20	0.7	5.78	94 415	8.92	1116	1663
S-9	2	0.2	0.1	20	0.9	11.78	78 907	7.49	994	1411
S-10	5	0.2	0.5	20	0.9	9.86	82 111	7.85	1016	1468
S-11	2	0.5	0.5	30	0.7	6.39	91 855	8.78	1110	1641
S-12	5	0.2	0.5	30	0.7	6.43	91 361	8.65	1109	1638
S-13	10	0.2	0.2	30	0.7	6.13	93 361	8.8	1110	1649
S-14	2	0.1	0.1	30	0.9	11.93	77 575	7.33	991	1390
S-15	2	0.2	0.5	30	0.9	12.14	77 139	7.26	976	1378
S-16	5	0.2	0.2	30	0.9	11.1	79 147	7.54	998	1442
S-17	5	0.5	0.1	30	0.9	11.06	79 259	7.55	1004	1447
S-18	10	0.1	0.5	30	0.9	11.15	79 069	7.53	995	1435
S-19	5	0.5	0.2	40	0.7	6.8	90 020	8.54	1097	1608
S-20	10	0.2	0.1	40	0.7	6.71	90 031	8.57	1104	1621
S-21	10	0.5	0	40	0.7	6.92	89 836	8.51	1081	1599
S-22	10	0.5	0.1	40	0.7	6.56	91 020	8.63	1108	1627
S-23	5	0.1	0.1	40	0.9	12.8	75 878	7.09	961	1363
S-24	5	0.1	0.5	40	0.9	12.09	77 226	7.28	978	1383
S-25	5	0.2	0	40	0.9	14.02	73 986	6.75	910	1321
S-26	10	0.1	0.5	40	0.9	12.72	76 011	7.16	964	1367
S-27	10	0.2	0.1	40	0.9	12.87	75 701	7.01	950	1353
S-28	10	0.5	0.1	40	0.9	12.83	75 815	7.05	957	1360
S-29	10	0.5	0.2	40	0.9	12.85	75 785	7.03	954	1356
S-30	10	0.5	0.5	40	0.9	12.78	75 902	7.12	962	1364
S-31	10	0.1	0.2	50	0.7	6.69	90 220	8.6	1105	1623
S-32	10	0.5	0.2	50	0.7	7.48	89 777	8.45	1068	1597
S-33	5	0.5	0	50	0.9	14.46	73 459	6.57	893	1308
S-34	10	0.2	0.5	50	0.9	13.88	74 215	6.86	924	1331
S-35	10	0.5	0.5	50	0.9	13.91	74 112	6.83	921	1328
S-6	10	0	0.5	50	0.9	13.86	74 312	6.88	928	1336

Table 4

Simulation parameters for performance comparison.

Parameter	Value
Network topology	Abilene, US26, SprintLink, Edge
C	5000
$ CS(R_x) $	30
$L(C_i)$	4
α	0.9
γ	0.1
$MaxSize(HQ(R_x))$	500
(β, θ, η)	5, 0.2, 0.2 (for Abilene-based topology)
(β, θ, η)	2, 0.2, 0.5 (for US26-based topology)
(β, θ, η)	2, 0.1, 0.2 (for SprintLink backbone topology)
(β, θ, η)	5, 0.2, 0.0 (for Edge network-based topology)
λ_{max}	10/second (for Abilene and US26) and 100/second (for SprintLink and Edge-topology)
Delay of links	$10 \times 10^3 \mu s$ (for Abilene and US26) and $10 \times 10^3 - 10 \times 10^5 \mu s$ (for SprintLink and Edge-topology)
Bandwidth of links	10Mbps (for Abilene and US26) and 100Kbps-10Mbps (for SprintLink and Edge-topology)

freshness and hop count parameters into caching decisions, fails to improve the hit ratio sufficiently. In this simulation scenario, the proposed PFC caching scheme outperformed its peer caching schemes by achieving a cache hit ratio gain up to 6.5%. During simulations on US26-based IoT network topology, the Consumer-Driven and Consumer-Cache approaches exhibit the lowest hit ratios among the caching strategies due to their strict dependency on content freshness and network topology. In this scenario, the MACD and the proposed PFC schemes obtain a higher cache hit ratio than the peer schemes. This highlights the advantage of adaptive and probabilistic caching mechanisms in environments with dynamic content demands. However, the proposed PFC approach surpasses the MACD scheme by enhancing the cache hit ratio by 5.1% from the MACD caching scheme. The superior performance of PFC highlights the importance of dynamically adjusting caching thresholds based on content popularity, its freshness, and hop count parameters.

During simulations on the SprintLink Core-network and the Edge-network topologies, the request rate has been increased to 100 requests per second, and dynamic link bandwidth has been applied to obtain realistic results. The optimal values for the coefficients β, θ , and η have been determined through multiple simulations, following the methodology used on Abilene and

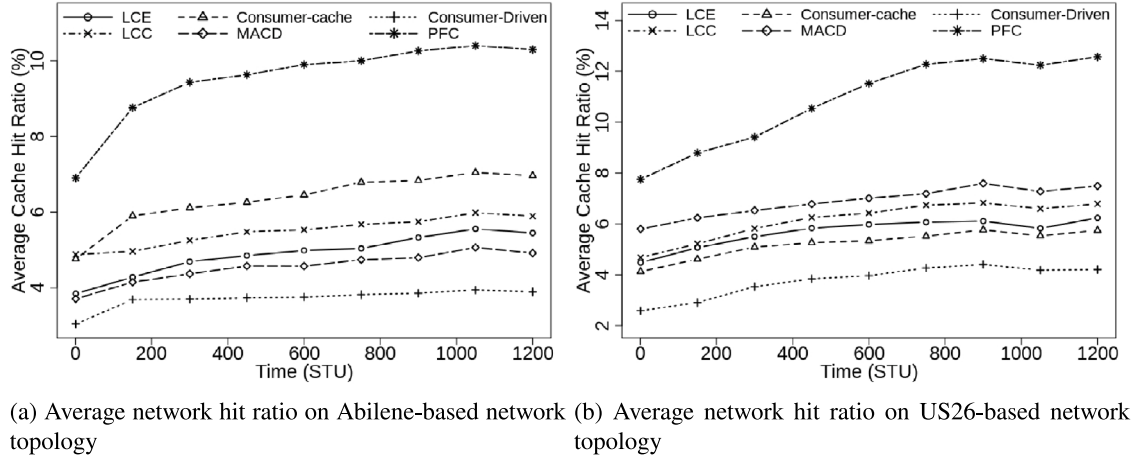


Fig. 6. In-network cache hit-ratio of caching schemes on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 30$, $\alpha = 0.9$ and $\lambda = 10/\text{second}$.

Table 5

In-network average cache hit-ratio of caching schemes on SprintLink and Edge network-based topologies.

Content caching strategy	Cache hit ratio on SprintLink network-based topology	Cache hit ratio on Edge network-based topology
LCE	$5.42\% \pm 0.88$	$16.89\% \pm 1.79$
Consumer Cache	$5.13\% \pm 0.90$	$16.89\% \pm 1.79$
Consumer-Driven	$4.31\% \pm 0.61$	14.48 ± 1.73
LCC	$4.39\% \pm 0.79$	$16.85\% \pm 1.81$
MACD	$5.74\% \pm 1.02$	$19.63\% \pm 1.81$
APDC	$5.56\% \pm 0.83$	$16.89\% \pm 1.79$
PFC	$10.62\% \pm 0.89$	$29.20\% \pm 2.59$

US26-based topologies. The value for simulator parameters are summarized in Table 4. Table 5 illustrates that the proposed PFC strategy outperforms the peer caching techniques by achieving a higher cache hit ratio on both, SprintLink and Edge network-based topologies. The results present the average value of the cache hit ratio metric with standard deviation values calculated between 800 and 1200 STU to reflect stabilized results after initial network convergence. On the SprintLink core-network topology, the PFC scheme achieved an impressive average network hit ratio of 10.62%, nearly doubling the performance of established techniques such as LCE (5.42%) and surpassing the MACD scheme (5.74%). This substantial improvement directly correlates with faster content retrieval and enhanced data availability within the network. The evaluation on Edge-network topology further illustrates the adaptability and superior cache hit ratio of the PFC scheme (29.20%) as shown in Table 5. Thus, these results underscore the suitability of the PFC scheme for deployment across stub, core, and edge network environments, where efficient resource utilization and rapid content delivery are essential.

5.2.2. Performance comparison: average network delay

The network delay for each Interest message is computed as the time elapsed (in micro-seconds) between generating the Interest message and receiving the corresponding Content message within the network. If an Interest message is lost in the IoT network, the delay is determined as the time between the generation of the first Interest message in the network and the subsequent reception of the corresponding content. This comprehensive metric encompasses various components, including transmission, propagation, queuing, and processing delays. The average network delay has been determined using Eq. (9) as given below.

$$Delay_{avg} = \frac{1}{n} \sum_{i=1}^n Sum(T_i, P_i, Q_i, Pr_i) \quad (9)$$

Here, n represents the total number of Interest messages generated in every STU for which the users have received the Content message. The parameters T_i , P_i , Q_i , and Pr_i denote total transmission delay, propagation delay, queuing delay, and processing delay experienced by the Interest message and its corresponding Content message in the network.

Fig. 7 and Table 6 show the average network delay experienced by end-users while retrieving IoT content on different network topologies. As illustrated in Fig. 7, the average network delay for all the caching schemes is high, between 0 and 200 STU (Simulation Time Unit) on the Abilene and US26-based network topologies. This initial surge in delay is observed due to the empty cache state of network routers, resulting in a higher number of cache misses that subsequently increase the delay in IoT content retrieval. However, as the routers begin to cache the contents, the delay tends to decrease due to the in-network caching of IoT contents. On Abilene network topology, the proposed PFC approach shows a 15.3%, 9.1%, 21.7%, 12.7%, and 17.7% reduction in average network delay from the LCE, Consumer-cache, Consumer-Driven, LCC, and MACD caching schemes respectively Fig. 7(a). The PFC method

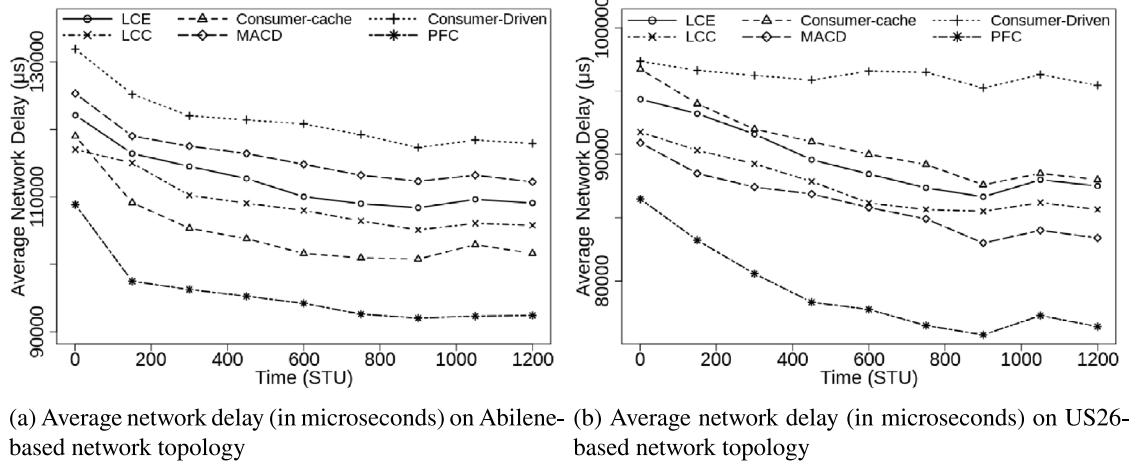


Fig. 7. Average network delay (in microseconds) of caching schemes on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 30$, $\alpha = 0.9$ and $\lambda = 10/\text{second}$.

Table 6

Average network delay (μ s) of caching strategies on SprintLink and Edge network-based topologies.

Content Caching Strategy	Network delay on SprintLink network-based topology	Network delay on Edge network-based topology
LCE	85913 $\pm$ 2376	36803 $\pm$ 363
Consumer Cache	86641 $\pm$ 2411	36803 $\pm$ 363
Consumer-Driven	88696 $\pm$ 1434	37171 $\pm$ 334
LCC	88291 $\pm$ 2319	36822 $\pm$ 368
MACD	84523 $\pm$ 2534	36399 $\pm$ 357
APDC	85641 $\pm$ 2213	36803 $\pm$ 363
PFC	75652 $\pm$ 2394	34425 $\pm$ 483

consistently decreases retrieval delay, from 108,830 μ s at STU = 1 to 92,400 μ s at STU = 1200. This 15.1% reduction shows PFC's ability to adapt its caching decisions to bring frequently requested information closer to requesting users. A similar reduction trend is also observed on the US26-based network topology, as depicted in Fig. 7(b). PFC has a delay of 86472 μ s at STU = 1, 5.5% lower than the next best scheme (MACD at 90,905 μ s). PFC decreases delay to 76,400 μ s at $t = 1200$, reflecting a total reduction of 11.6%. All caching techniques demonstrate that the US26-based IoT network has faster retrieval times than the Abilene-based network topology. The US26 network has a reduced average path length due to its compactness. However, the performance advantage of PFC remains stable across topologies, demonstrating its flexibility.

Analogous to previous results (shown in Fig. 7), Table 6 presents the average network delay observed by various content caching strategies during content retrieval in SprintLink and Edge network-based topologies. In contrast to the compared approaches, the proposed PFC scheme demonstrates the lowest network delay on both topologies. The PFC scheme achieved a network delay of 75652 $\pm$ 2394 μ s on the SprintLink network and 34425 $\pm$ 483 μ s on the Edge network. These results prove the efficiency of the PFC scheme in reducing data transmission delays, making it a superior caching strategy for delay sensitive applications compared to conventional approaches like LCE, Consumer Cache, Consumer-Driven, LCC, MACD, and APDC.

5.2.3. Performance comparison: average network hop count and average network traffic

We have also conducted a comparative analysis of the proposed PFC caching scheme with existing strategies based on two critical metrics: average network hop count and average network traffic. The comparisons are illustrated in Figs. 8 and 9. The hop-count metric is obtained by counting the number of hops traversed by the Interest message and its corresponding Content message in the network as defined in Eq. (10).

$$\text{HopCount}_{avg} = \frac{1}{n} \sum_{i=1}^n \text{HopCount}_i \quad (10)$$

In this equation, HopCount_i is computed as the sum of the hop count observed by an Interest message “i” to reach the content provider and its matching Content message to reach the requesting user. Here, n represents the total number of Interest messages generated in per STU. A lower hop-count value signifies that the IoT content is accessed from nearby network routers. This, in turn, indicates reduced load on IoT devices in the network. As depicted in Figs. 8(a) and 8(b), our proposed caching scheme significantly reduces the average hop count during content access. On Abilene network topology, the PFC scheme achieves a 9.9%–17.8% reduction in hop count compared to the peer caching schemes. The average hop count for Consumer-cache strategy steadily drops before stabilizing at 10.35 at STU = 1200, which is significantly greater (10.2%) than the PFC. Analogous simulation results are observed on

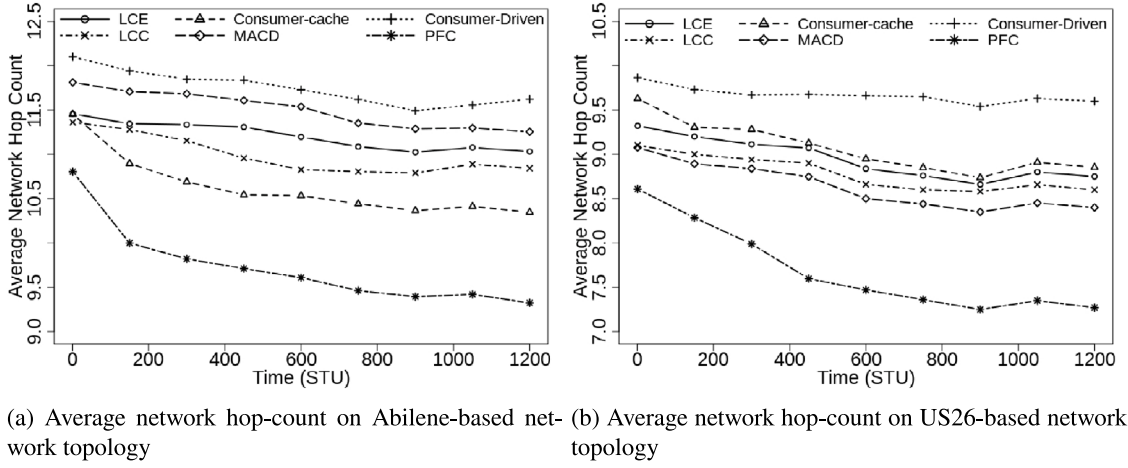


Fig. 8. Average network hop-count of caching schemes on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 30$, $\alpha = 0.9$ and $\lambda = 10/\text{second}$.

Table 7

Average network hop-count of caching strategies on SprintLink and Edge network-based topologies.

Content caching strategy	Hop-count on SprintLink network-based topology	Hop-count on Edge network-based topology
LCE	8.57 ± 0.24	3.69 ± 0.04
Consumer Cache	8.68 ± 0.26	3.69 ± 0.04
Consumer-Driven	8.96 ± 0.09	3.73 ± 0.03
LCC	8.87 ± 0.15	3.72 ± 0.04
MACD	8.38 ± 0.26	3.63 ± 0.05
APDC	8.53 ± 0.18	3.69 ± 0.04
PFC	7.56 ± 0.24	3.43 ± 0.05

US26-based topology, reinforcing the effectiveness of the proposed PFC scheme. Notably, the PFC scheme shows improved network performance from the existing caching strategies, as illustrated in 8(b).

The simulation results for SprintLink and Edge network-based topologies further demonstrate the superiority of the proposed PFC scheme. As shown in Table 7, PFC achieves the lowest average hop count on both network topologies. In the SprintLink network, traditional methods such as LCE and Consumer Cache exhibit higher hop counts of 8.57 ± 0.24 and 8.68 ± 0.26 , respectively. This is much higher than the obtained 7.56 ± 0.24 hop count by PFC. Similar patterns are observed on the Edge topology, where peer strategies demonstrate higher hop counts than the proposed PFC strategy.

The average network traffic metric is computed as the traffic load in kilobytes per second (KB/s) across all communication links over a specific time interval as shown in Eq. (11).

$$T_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^{M_i} L_{ij} \quad (11)$$

where, T_{avg} , N , M_i , and L_{ij} represents average network traffic (KB/s), number of communication links, number of packets transmitted over link i in per unit time, and size of the j -th packet transmitted over link i (in KB), respectively.

A reduction in network traffic signifies more efficient utilization of network bandwidth, with popular and fresh IoT contents strategically cached closer to the end-users. Fig. 9 plots the average network traffic observed by various caching schemes on Abilene and US26-based IoT networks. Earlier, the PFC caching scheme was shown to observe a lesser delay in retrieving the IoT contents. Therefore, the proposed scheme also decreases the load on communication links. On Abilene-based IoT network, the PFC scheme reduces the load on network links by 15.3%, 10.5%, 17.9%, 13.5%, and 16.4% as compared to the LCE, Consumer-cache, Consumer-Driven, LCC, and MACD caching schemes respectively (Fig. 9(a)). Similarly, during simulations on the US26-based IoT topology, the PFC approach consistently outperforms its peer caching schemes by reducing network traffic within the range of 8.5% – 21.1% compared to existing caching strategies Fig. 9(b). It has been observed that US26 has less network traffic than Abilene-based network topology. This is due to a smaller user base (requesters), which means fewer content requests and lower traffic volumes. However, PFC continues to reduce network traffic better than peer strategies, supporting performance trends aligned with Abilene network.

The network traffic results observed on SprintLink and Edge network-based topologies are presented in Table 8. In both networks, existing strategies such as LCE, Consumer Cache, and Consumer-Driven experience higher traffic loads than the PFC scheme, indicating inefficient bandwidth utilization. The reduction in traffic highlights the ability of the PFC scheme to alleviate congestion on communication links by efficiently caching content at optimal locations and thus, results in enhanced utilization of network resources.

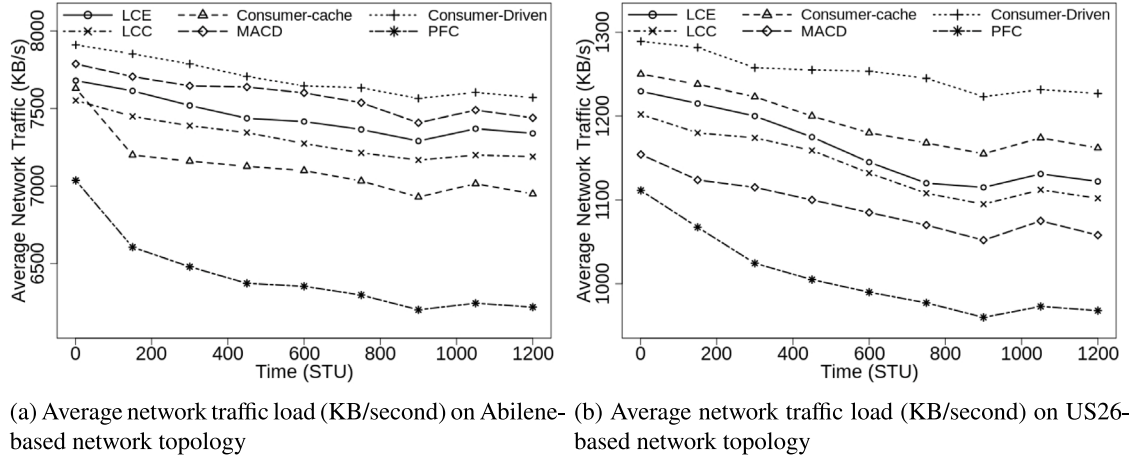


Fig. 9. Average network traffic load (KB/second) of caching schemes on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 30$, $\alpha = 0.9$ and $\lambda = 10/\text{second}$.

Table 8

Average network traffic load (KB/second) of caching strategies on SprintLink and Edge network-based topologies.

Content caching strategy	Traffic on SprintLink network-based topology	Traffic on Edge network-based topology
LCE	1725 ± 89	1877 ± 17
Consumer Cache	1748 ± 96	1877 ± 17
Consumer-Driven	1774 ± 64	1895 ± 13
LCC	1765 ± 83	1881 ± 18
MACD	1696 ± 95	1838 ± 19
APDC	1713 ± 86	1877 ± 17
PFC	1555 ± 93	1767 ± 21

5.2.4. Performance comparison: average energy consumption

In Fig. 10, we observe average, minimum, and maximum energy consumption patterns within IoT networks. The total energy consumption metric evaluates the efficiency of content caching approaches. An optimized caching approach reduces the load from IoT devices and significantly reduces the energy requirements. By accessing IoT-devices less frequently, we conserve their energy resources. The total energy consumed by IoT devices (both for sensing and transmitting content) and routers is calculated using Eq. (12) [54] as follows:

$$E_{total} = \sum_{i=1}^{|q|} n_i \times [(e_i + e_{sense})] + \sum_{j=1}^{|r|} n_j \times [e_j] \quad (12)$$

Here, $|q|$ and $|r|$ represent the total number of routers and IoT devices in the IoT network. The variables n_i and n_j define the number of Interest messages received by each IoT device and network router. The energy IoT devices use to sense the content is represented by e_{sense} . The energy IoT devices and routers used to forward the content are represented by e_i and e_j , respectively. Generally, the $[(e_i + e_{sense})] \gg e_j$ suggests that sensing the content consumes more energy than forwarding the content. It has been assumed that $[(e_i + e_{sense})]$ needs 4 millijoule energy and the $[e_j]$ requires 1 millijoule energy during simulations.

On both IoT networks (Abilene and US26-based), the proposed PFC caching scheme significantly reduces energy consumption compared to existing strategies. While the LCE caching scheme increases content redundancy in the network, the scheme outperforms the MACD and Consumer-Driven caching schemes on Abilene-based IoT network topology. This is because both caching systems assign a low caching probability to less fresh contents, even when network routers have available cache space. Fig. 10(a) illustrates that the proposed PFC approach shows a 9.7%–14.4% reduction in average energy consumption from existing schemes. On the US26-based IoT network topology, all the caching schemes exhibit reduced energy consumption due to fewer network nodes. Here, the PFC strategy reduces the energy consumption up to 20.8% as compared to peer approaches, as shown in Fig. 10(b).

Table 9 presents the average energy consumption in the SprintLink and Edge network-based topologies. Under SprintLink topology, traditional caching strategy LCE and Consumer-Driven approach exhibit higher energy consumption, which is 2602 ± 105 and 2663 ± 65 , respectively. This is due to inefficient caching placement policies that trigger excessive sensing and transmission operations in the network. In this topology, PFC reduces the frequency of content sensing and enhances forwarding operations at the router level. A similar reduction in energy consumption has been observed by the proposed PFC approach in the Edge network-based topology. This superior performance of PFC effectively meets the energy demands of modern IoT networks due to its underlying caching mechanism, which minimizes redundant data transmission and places contents closer to the end-users for a longer duration.

The simulation results shown in Table 10 highlight the performance improvement achieved by the PFC caching scheme compared to our previous work, APDC strategy, on different network topologies. The PFC scheme consistently outperforms APDC in terms of

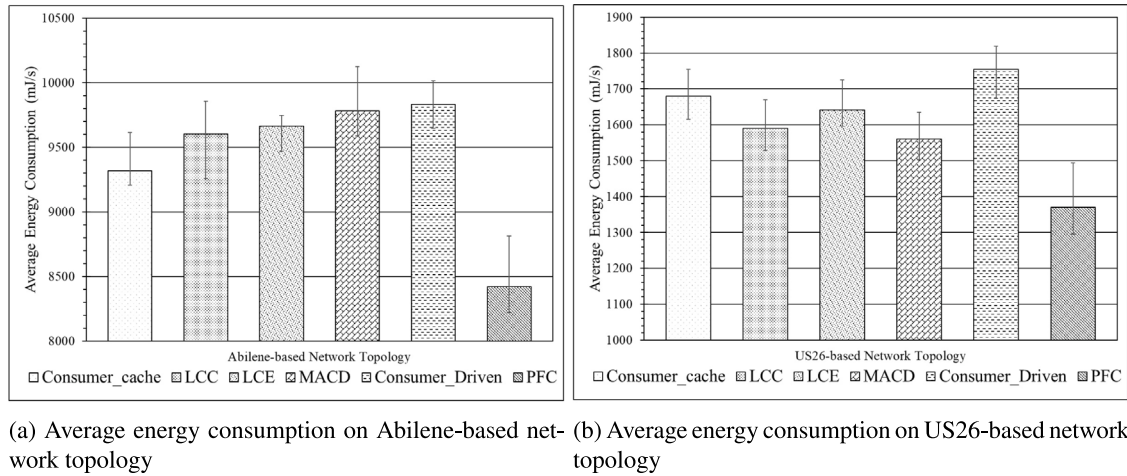


Fig. 10. Average energy consumption (mJ/s) of caching schemes on Abilene and US26-based topology with $|D| = 5000$, $|CS(FN_i)| = 30$, $\alpha = 0.9$ and $\lambda = 10/\text{second}$.

Table 9

Average energy consumption (mJ/s) of caching strategies on SprintLink and Edge network-based topologies.

Content caching strategy	Energy consumption on SprintLink network-based topology	Energy consumption on Edge network-based topology
LCE	2602 ± 105	4185 ± 72
Consumer Cache	2621 ± 108	4185 ± 72
Consumer-Driven	2662 ± 65	4252 ± 64
LCC	2656 ± 86	4189 ± 71
MACD	2568 ± 127	4094 ± 76
APDC	2584 ± 92	4185 ± 72
PFC	2289 ± 106	3773 ± 87

Table 10

Performance comparison of APDC and PFC strategies on different network topologies.

Network topology	Caching strategy	Hit ratio (%)	Delay (μs)	Hop count	Traffic (KB/s)
Abilene-based	APDC	5.36 ± 0.35	108067.99 ± 1441.58	11.04 ± 0.17	7281 ± 251
	PFC	10.22 ± 0.21	92698 ± 865.74	9.37 ± 0.11	6224 ± 62
US26-based	APDC	6.59 ± 1.13	86631.08 ± 2716.40	8.65 ± 0.28	1090 ± 57
	PFC	12.14 ± 0.42	77139 ± 1019.32	7.26 ± 0.13	976 ± 11
Core-network	APDC	5.56 ± 0.83	85641 ± 2213	8.53 ± 0.18	1713 ± 86
	PFC	10.62 ± 0.89	75652 ± 2394	7.56 ± 0.24	1555 ± 93
Edge-network	APDC	16.89 ± 1.79	36803 ± 363	3.69 ± 0.04	1877 ± 17
	PFC	29.20 ± 2.59	34425 ± 483	3.43 ± 0.05	1767 ± 21

higher average network hit ratio and reduced average network delay, hop count, and network traffic. Notably, in the edge topology, PFC achieves a 29.20% hit ratio, significantly surpassing the hit ratio of APDC (16.89%). This performance enhancement is observed due to consideration of content freshness, popularity, and hop count during caching decisions, while the APDC strategy could be more suitable for static content that does not expire over time. As we focus on managing transient content in IoT networks, PFC places popular and fresh content close to the users, making it a more effective and scalable solution in NDN-based IoT environments.

The extensive simulation executions and performance comparisons reveal that the existing peer content caching approaches have not taken optimal caching decisions. In contrast, the proposed PFC caching strategy takes a dynamic approach that does not set a constant or global caching threshold for all routers in the IoT network. Instead, the approach adaptively adjusts the threshold based on the content freshness and the distance-based parameters, ensuring efficient utilization of cache resources. During caching decisions, PFC compares the popularity of transient content with the computed threshold. The caching probability increases with the content popularity, freshness value, and the distance traversed by the content. Due to this dynamic content placement decision heuristic, the scheme possesses high adaptability and seamlessly fits into various IoT networks, accommodating realistic content access patterns. Thus, changes in network topology would have a negligible impact on the caching performance of the proposed approach.

6. Conclusion

In this paper, we proposed a novel Popularity and Freshness-based Caching (PFC) scheme for NDN-based IoT networks. The PFC scheme strategically caches popular and fresh contents near end-user devices, addressing the challenges associated with the transient nature of IoT contents. The routers make autonomous caching decisions based on three key factors: content access frequency, content freshness, and router proximity. To determine the popularity of contents, the PFC approach implements a request history queue on each network router. By using this information and the underlying heuristics, the scheme increases the probability of retrieving popular and fresh content from nearby routers, while the less frequently accessed content is obtained directly from IoT devices. This mechanism ensures that crucial IoT contents are cached on the network routers without introducing significant network overhead. Comprehensive simulations are performed on Abilene, US26, SprintLink, and Edge network-based IoT topologies, and the simulation results are compared with existing state-of-the-art caching strategies. The results showed that the PFC scheme achieved significant improvements across various performance metrics, including cache hit ratio, delay, network traffic, and energy consumption. This makes the proposed caching solution suitable for enhancing network performance in large-scale IoT applications.

In future work, we will investigate the impact of end-user mobility on the PFC caching solution. Furthermore, we will increase the accuracy of caching decisions by analyzing the integration of other characteristics during content caching.

CRedit authorship contribution statement

Sumit Kumar: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rajeev Tiwari:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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